project	Nonetheless, this does not preclude submission and selection of a proposal requesting different amounts.
Indicative budget	The total indicative budget for the topic is EUR 1.00 million.
Type of Action	Coordination and Support Actions

Expected Outcome: Projects are expected to contribute to the following expected outcomes:

- A mapping of the forthcoming technology process trends and the IP framework relevant for R&I that will go with these
- An assessment of the current intellectual property methods and how fit for purpose they are taking into account the transformation of research results into markets / socio-economic benefits
- Development of scenarios with potential new protection mechanisms that will benefit both the research community and society.
- Set of guidelines to R&I actors, notably the research generators, for the protection and use of intellectual property assets.
- Set of recommendations for policy makers regarding the future needs for framework conditions relevant for R&I.
- Establishment of a communication channel with the aim of promoting an open dialogue on R&I driven intellectual property needs, bringing together various stakeholders and practitioners and providing a feedback loop between practitioners and policy makers regarding emerging needs.

Scope: Technology sovereignty requires, together with investment, an in-depth understanding of the research and innovation landscape. The current Open Innovation strategies, the ever growing Open Science practices and the artificial intelligence revolution create an ecosystem where the innovations happen faster than ever (machine learning, big data etc.).This topic will look at ways to overcome the existing gaps researchers have to face when following unconventional discovery methods (e.g. artificial intelligence, computer inventions etc.).

DEEPENING THE EUROPEAN RESEARCH AREA

OPEN SCIENCE

Proposals are invited against the following topic(s):

HORIZON-WIDERA-2021-ERA-01-40: Modelling and quantifying the impacts of open science practice

Specific conditions

Expected EU contribution per project	The Commission estimates that an EU contribution of around EUR 2.00 million would allow these outcomes to be addressed appropriately. Nonetheless, this does not preclude submission and selection of a proposal requesting different amounts.
Indicative budget	The total indicative budget for the topic is EUR 2.00 million.
Type of Action	Research and Innovation Actions
Legal and financial set-up of the Grant Agreements	The rules are described in General Annex G. The following exceptions apply: Beneficiaries will be subject to the following additional obligations regarding open science practices: Beneficiaries must ensure early and open sharing of the strategies, methodologies, models, and raw and analysed data deriving from their activities. Beneficiaries must acknowledge and incorporate these obligations in the proposal, outlining the efforts they will make towards meeting them and in Annex I to the Grant Agreement.

Expected Outcome: Projects are expected to contribute to the following expected outcomes:

- Structured evidence of scientific, societal and economic impacts of open science practice, complemented by the new methods, tools and data required to measure them.
- Better knowledge on how impacts develop and how a diversity of benefits can be achieved at scientific, societal and economic levels.
- Better understanding on how the practice of open science can contribute to the increased reproducibility of research results.

These targeted outcomes in turn contribute to medium and long-term impacts:

- Prioritisation of policy actions for open science, and improved policy making on open science.
- Maximised impact of open science.
- Increased capacity in the EU R&I system to conduct open science and to set it as a *modus operandi* of modern science.

Scope: Open science consists of sharing knowledge and data as early as possible in the research process, in open collaboration with all relevant actors, including citizens. The mainstreaming of open science practice[43] is driven by expected impacts: (i) on the research

[43] Open science practice includes: providing open access to research outputs (such as publications, data, software, models, algorithms, and workflows); early and open sharing of research (for example through preregistration, registered reports, pre-prints, crowd-sourcing of solutions to a specific problem); participation in open peer-review; measures to ensure reproducibility of results; and involving citizens, civil society and end-users in the co-creation of R&I agendas and content, including citizen science.